

Week 5: Functions

BSC129 / BSC136

- Functions
- Domain, codomain, image, preimage
- Composition of functions
- Injections, surjections, bijections
- Inverse of functions
- Cardinality, (un-)countability

Definition

Let A and B be nonempty sets.

- A **function** f (also called **mappings** or **transformations**) from A to B is an assignment **from each** element of A **to exactly one** element of B .
- In terms of logic: if b is the element of B assigned by the function f to the element a of A , we write:

$$\forall a \in A, \exists! b \in B, \underline{f(a) = b.}$$

b depends on a via the function f

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- In terms of logic: if b is the element of B assigned by the function f to the element a of A , we write:

$$\forall a \in A, \exists! b \in B, f(a) = b.$$

- We say that f **maps** A to B .
- We write

$$f : A \rightarrow B, \quad f(a) = b$$

(or sometimes in short $f : a \mapsto b$).

Example.

- $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2.$

Example. $f(1.1) = 1.21 = f(-1.1), f(\pi) = \pi^2, \dots$

- $f : \mathbb{R}^2 \rightarrow \mathbb{R}, f(x, y) = x + y.$

Example. $f(1, 2) = 3, f(\pi, e) = \pi + e = f(e, \pi), \dots$

- Truth value

$$TV : \{\text{propositions}\} \rightarrow \{T, F\}.$$

$TV("2+2 = 4") = T$, while $TV("2+2 = 5") = F$.

- Predicate:

$$P : D \rightarrow \{\text{propositions}\}$$

Here D is a set (or a domain).

Example.

$D = \{\text{Mon, Tue, ..., Sat, Sun}\}$. $P(x) = \text{"It rains on } x\text{"}$.

- Power set:

$$\mathcal{P} : \{\text{sets}\} \rightarrow \{\text{sets of subsets}\}$$

such that $\mathcal{P}(S) = \{\text{subsets of } S\}$.

Example.

$S = \{1, 2, 3\}$, $\mathcal{P}(S) = \{\text{emptyset}, \{1\}, \{2\}, \{3\}, \{1,2\}, \{2,3\}, \{1,3\}, S\}$

Definition

- If f is a function from A to B , we say that

A is the **domain (source set)** of f , and
 B is the **codomain (target set)** of f .

- If $f(a) = b$, we say that

b is **the image** of a , and
 a is **a preimage** of b .

When the input a is given, the image $f(a)$ is the unique output.
Conversely, an image can come from multiple preimages.

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b is **the image** of a , and
 a is **a preimage** of b .

- The **image**, or **range**, of f is the subset in B of all images from elements of A , denoted by $f(A)$.

$$f(A) = \{f(a) | a \in A\} \subseteq B.$$

The image of f is a subset of the codomain of f , possibly a proper subset.

Example Let $f : \mathbb{N} \rightarrow \mathbb{N} \cup \{0\}$ be a function that assigns to every positive integer the last digit of that integer.

① Find $f(101)$, $f(1234)$, and $f(20)$. 1, 4, 0

② What is the domain of f ? $\mathbb{N}$

③ What is the codomain of f ? $\mathbb{N} \cup \{0\}$

④ What is the image of f ?

The possible digits are $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$, which form the image of f .

Example Let $f : \mathbb{N} \rightarrow \mathbb{N} \cup \{0\}$ be a function that assigns to every positive integer the last digit of that integer.

- 1 Find $f(101)$, $f(1234)$, and $f(20)$.
- 2 What is the domain of f ?
- 3 What is the codomain of f ?
- 4 What is the image of f ?

Example Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function that assigns to a real number the square of this number. $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2$.

- 1 Find $f(1.2)$ and $f(x)$. $1.44, x^2$
- 2 What is the domain of f ? $\mathbb{R}$
- 3 What is the codomain of f ? $\mathbb{R}$
- 4 What is the image of f ? $[0, \text{infinity})$

We can define addition and multiplication of real-valued or integer-valued functions.
 (codomain is $\mathbb{R}$ for real-valued, codomain is $\mathbb{Z}$ for integer-valued)

Example. Let f_1 and f_2 be functions from $\mathbb{R}$ to $\mathbb{R}$, $f_1(x) = x$ and $f_2(x) = x - x^3$. Find $g : \mathbb{R} \rightarrow \mathbb{R}$ where

1 $g = f_1 + f_2$ $g(x) = f_1(x) + f_2(x) = x + (x - x^3) = 2x - x^3.$

2 $g = f_1 - f_2$ $g(x) = f_1(x) - f_2(x) = x - (x - x^3) = x^3$

3 $g = f_1 f_2$ $g(x) = f_1(x) f_2(x) = x(x - x^3) = x^2 - x^4$

We can define addition and multiplication of real-valued or integer-valued functions.

Example. Let f_1 and f_2 be functions from $\mathbb{R}$ to $\mathbb{R}$, $f_1(x) = x$ and $f_2(x) = x - x^3$. Find $g : \mathbb{R} \rightarrow \mathbb{R}$ where

① $g = f_1 + f_2$

② $g = f_1 - f_2$

③ $g = f_1 f_2$

Beware: $g = f_1/f_2$ is defined only on

$$\{x \in \mathbb{R} : f_2(x) \neq 0\}.$$

This is the new domain for f_1/f_2 , as to avoid dividing zero.

Example. Let f_1 and f_2 as above, then $g : \mathbb{R} \setminus \{0, 1, -1\} \rightarrow \mathbb{R}$.

Factorize $f_2(x) = x(1-x)(1+x)$, so $f_2(x) = 0$ exactly when $x = 0, 1, -1$.

Composition of functions

Condition: the codomain of f is equal to the domain of g

Given the functions $f : A \rightarrow B$ and $g : B \rightarrow C$, the **composition** of g and f , denoted by $g \circ f$, is a function from A to C given by

$$(g \circ f)(a) = g(f(a)).$$

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Example. Given

$$f : \mathbb{Z} \rightarrow \mathbb{Z}, f(n) = 2n + 3,$$

$$g : \mathbb{Z} \rightarrow \mathbb{Z}, g(n) = 3n + 2,$$

find $g \circ f$ and $f \circ g$. $(g \circ f)(n) = g(f(n)) = g(2n+3) = 3(2n+3) + 2 = 6n + 11.$
 $(f \circ g)(n) = f(g(n)) = f(3n+2) = 2(3n+2) + 3 = 6n + 7.$

Question. Is $g \circ f = f \circ g$? **No in general.**

The Graphs of Functions

Given the function $f : A \rightarrow B$, the graph of f is the set

$$G_f = \{(a, b) \mid a \in A \text{ and } b = f(a)\} \subseteq A \times B.$$

(a,b) is an element in the Cartesian product $A \times B$.

We can also write $G_f = \{(a, f(a)) \mid a \in A\}$.

The Graphs of Functions

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Example. Given

$$f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2 - 1,$$

find the graph of f . $G_f = \{(x, x^2 - 1) \mid x \in \mathbb{R}\}$

Pictorial graph: (refer to the whiteboard)

The Graphs of Functions

Given the function $f : A \rightarrow B$, the graph of f is the set

$$G_f = \{(a, b) \mid a \in A \text{ and } b = f(a)\} \subseteq A \times B.$$

Example. Given

$$f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2 - 1,$$

find the graph of f .

Example. Given

$$g : \{-2, -1, 0, 1, 2, 3, 4\} \rightarrow \mathbb{Z}, g(x) = x^2 - 1,$$

find the graph of g .

$$G_g = \{(x, x^2 - 1) \mid x \in \{-2, -1, 0, 1, 2, 3, 4\}\}$$

$$= \{(-2, 3), (-1, 0), (0, -1), (1, 0), (2, 3), (3, 8), (4, 15)\}$$

Pictorial graph of g : (refer to the seven points on whiteboard)



The floor function

The floor function assigns to the real number x the largest integer that is less than or equal to x . It is denoted by $\lfloor x \rfloor$.

Example

• $\lfloor \frac{1}{2} \rfloor = 0$, while $\lfloor -\frac{1}{2} \rfloor = -1$

 0.5 -0.5

$\text{floor}(-1.2) = -2$
 $\text{floor}(-7.2345) = -8.$

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- $\lfloor 2 \rfloor = 2$, and $\lfloor -2 \rfloor = -2$

The floor function

The floor function assigns to the real number x the largest integer that is less than or equal to x . It is denoted by $\lfloor x \rfloor$.

Example

- $\lfloor \frac{1}{2} \rfloor =$, while $\lfloor -\frac{1}{2} \rfloor =$
- $\lfloor 2 \rfloor =$, and $\lfloor -2 \rfloor =$

Example. Alina has 134 eggs to make cakes. If each cake needs 5 eggs, how many cakes can she bake?

$$134 / 5 = 26.8$$

She can make $\text{floor}(26.8) = 26$ cakes.

The ceiling function

The ceiling function assigns to the real number x the smallest integer that is greater than or equal to x . It is denoted by $\lceil x \rceil$.

- $\lceil \frac{1}{2} \rceil = 1$, while $\lceil -\frac{1}{2} \rceil = 0$
0.5 -0.5

The ceiling function

The ceiling function assigns to the real number x the smallest integer that is greater than or equal to x . It is denoted by $\lceil x \rceil$.

- $\lceil \frac{1}{2} \rceil = 1$, while $\lceil -\frac{1}{2} \rceil = 0$
- $\lceil 2 \rceil = 2$, and $\lceil -2 \rceil = -2$

The ceiling function

The ceiling function assigns to the real number x the smallest integer that is greater than or equal to x . It is denoted by $\lceil x \rceil$.

- $\lceil \frac{1}{2} \rceil = 1$, while $\lceil -\frac{1}{2} \rceil = 0$
- $\lceil 2 \rceil = 2$, and $\lceil -2 \rceil = -1$

Example. There are 379 books to be packed into boxes. If each box can only carry 8 books, at least how many boxes are required?

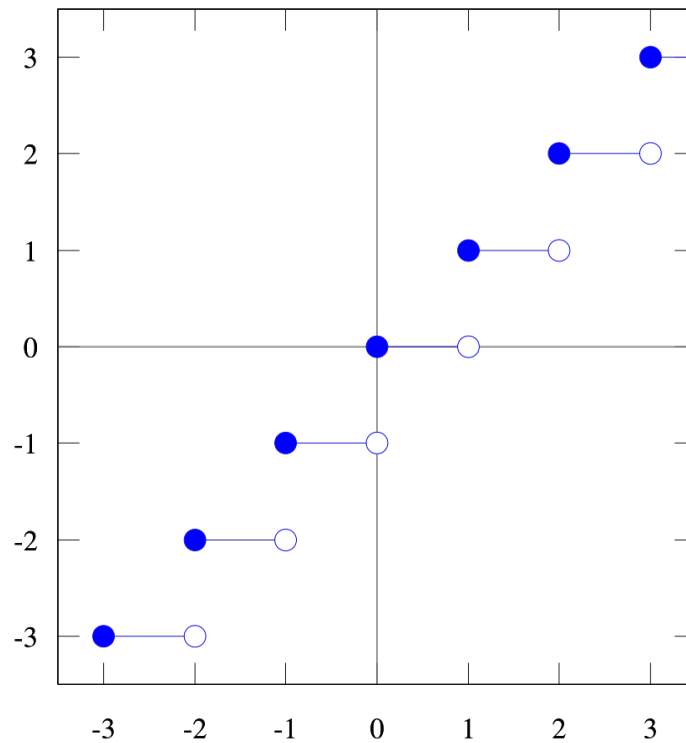
$$379 / 8 = 47.375$$

We need $\text{ceiling}(47.375) = 48$ boxes.

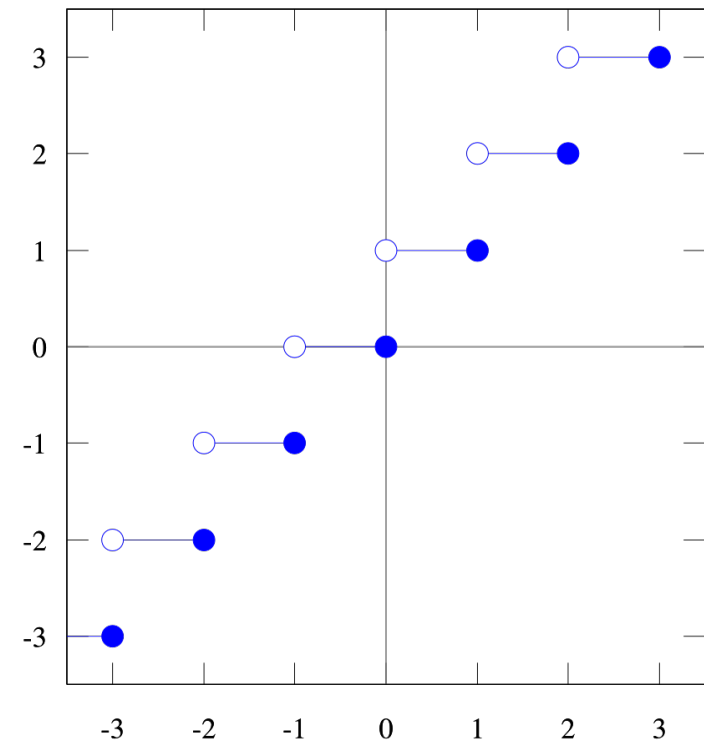
(We still need one box to carry the 3 remaining books.)

Graphs of floor and ceiling functions

Floor function



Ceiling function



Given a function $f: A \rightarrow B$, the image of f is the set $f(A) = \{ f(a) \mid a \text{ in } A \}$

Fix an element a in A , then the image of a via f is $f(a)$.

Definition

If f is a function from A to B and S is a subset of A , the **image** of S under f is

$$f(S) = \{ t \in B \mid \exists s \in S, t = f(s) \}.$$

We can also write $f(S) = \{ f(s) \mid s \text{ in } S \}$.

Definition

If f is a function from A to B and S is a subset of A , the **image** of S under f is

$$f(S) = \{t \in B \mid \exists s \in S, t = f(s)\}.$$

Example. Let $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2 - 1$. Given

$$S = \{n \in \mathbb{Z} \mid |n| \leq 3\} = \{-3, -2, -1, 0, 1, 2, 3\},$$

$$T = (-1, 2) = \{x \in \mathbb{R} \mid -1 < x < 2\},$$

find $f(S)$ and $f(T)$.

$$f(S) = \{f(-3), f(-2), \dots, f(3)\} = \{8, 3, 0, -1, 0, 3, 8\} = \{-1, 0, 3, 8\}.$$

$$f(T) = f((-1, 2)) = (-1, 3) \text{ (need to draw the graph to observe)}$$

Definition

If f is a function from A to B and T is a subset of B , the **preimage** of T under f is

$$f^{-1}(T) = \{a \in A \mid f(a) \in T\}.$$

Solve the equation to find all a in A such that $f(a)$ in T .

Example. Let $f : \mathbb{Z} \rightarrow \mathbb{Z}$, $f(x) = x^2 - 1$. Given

$$S = \{n \in \mathbb{N} \mid n \leq 10\} = \{1, 2, \dots, 9, 10\}, \text{ subset of } B$$

$$T = [-5, 20] = \{x \in \mathbb{R} \mid -5 \leq x \leq 20\},$$

find $f^{-1}(S)$ and $f^{-1}(T)$.

$f^{-1}(S) = \{a \in \mathbb{Z} \mid f(a) = \underline{a^2 - 1 \leq 10}\}$. Solving $a^2 \leq 11$, we get $a = \{-3, -2, -1, \dots, 3\}$.
solve for a So $f^{-1}(S) = \{-3, -2, -1, \dots, 3\}$.

To find $f^{-1}(T)$, we solve $-5 \leq a^2 - 1 \leq 20$, or $-4 \leq a^2 \leq 21$, or $0 \leq a^2 \leq 16$

$$f^{-1}(T) = \{-4, -3, -2, -1, \dots, 3, 4\}.$$

Towards bijective functions

Bijjective functions are also called invertible functions.

Recall: What is the criteria to define a function?

A **function** $f : A \rightarrow B$ is an assignment **from each** element of A **to a unique** element of B ($\forall a \in A, \exists! b \in B, f(a) = b$).

Towards bijective functions

Recall: What is the criteria to define a function?

A **function** $f : A \rightarrow B$ is an assignment **from each** element of A **to a unique** element of B ($\forall a \in A, \exists! b \in B, f(a) = b$).

- “**from each** element of A ...” means every element in A can be mapped to an element in B :

$$\forall a \in A, \exists b \in B, f(a) = b.$$

- “**to a unique** element of B .” can be written as:

$$\forall a_1, a_2 \in A, a_1 = a_2 \Rightarrow f(a_1) = f(a_2).$$

We want to define a “function” that is in the reversed direction of f , so this “function” $g: B \rightarrow A$ should also satisfy the above two criteria for functions.

Example. $f : \mathbb{R} \rightarrow \mathbb{R}, f(r) = (\text{solution}(s) \text{ of } x^2 = r).$

(spoiler: f is not a function !!!)

The function $e: \mathbb{R} \rightarrow \mathbb{R}, e(x) = x^2$. Question: Can we define the “inverse” of e ?
Intuitively, e is the square function, so the inverse should be the “square-root function”.
But there are two square roots (positive and negative), or even does not exist !!

Check the two criteria for defining functions:

(1) $\forall a \in \mathbb{R}$, can we find $b \in \mathbb{R}$ such that $f(a) = b$, that means can we solve $b^2 = a$ for b ?
When a is negative, there is no solution for b . So the answer is No.

(2) $\forall a_1, a_2 \in \mathbb{R}$, Is it true that $a_1 = a_2 \Rightarrow f(a_1) = f(a_2)$? If we take $a_1 = a_2 = 2$, and we assign $f(a_1) = \sqrt{2}$ but $f(a_2) = -\sqrt{2}$, then $f(a_1)$ is not equal to $f(a_2)$.
So the answer is also No.

Both criteria are violated. We cannot define “the inverse of the square”-function.

If we want to define the ‘inverse’ of the function f ,

$$g : B \rightarrow A, \quad g(b) = a \quad \Leftrightarrow \quad f(a) = b.$$

We don’t know if g exists or not. If g exists, then g **should also be a function**, so satisfies the two criteria:

If we want to define the ‘inverse’ of the function f ,

$$g : B \rightarrow A, \quad g(b) = a \quad \Leftrightarrow \quad f(a) = b.$$

We don’t know if g exists or not. If g exists, then g **should also be a function**, so satisfies the two criteria:

- “from each element of B ...”:

$$\forall b \in B, \exists a \in A, g(b) = a.$$

We don’t have g yet, but we have f already, so we rewrite:

$$\boxed{\forall b \in B, \exists a \in A, f(a) = b.}$$

The first (surjective) criterion for the existence of the inverse of f .

If we want to define the ‘inverse’ of the function f ,

$$g : B \rightarrow A, \quad g(b) = a \quad \Leftrightarrow \quad f(a) = b.$$

We don’t know if g exists or not. If g exists, then g **should also be a function**, so satisfies the two criteria:

- “from each element of B ...”:

$$\forall b \in B, \exists a \in A, g(b) = a.$$

We don’t have g yet, but we have f already, so we rewrite:

$$\forall b \in B, \exists a \in A, f(a) = b.$$

- “... to exactly one element of a .”:

$$\forall b_1, b_2 \in B, b_1 = b_2 \Rightarrow g(b_1) = g(b_2).$$

Again, we don’t have g yet, but we have f already, so we rewrite:

The second (injective) criterion for the existence of the inverse of f .

$$\forall a_1, a_2 \in A, f(a_1) = f(a_2) \Rightarrow a_1 = a_2.$$

Definition

A function f is said to be **injective** (or **one-to-one** or an **injection**) if

$$f(a_1) = f(a_2) \text{ implies } a_1 = a_2.$$

Example. $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$ is not injective, because $f(a_1) = f(a_2) = a^2$.
When a is not 0, the above does not imply $a_1 = a_2$: we can have $a_1 = a$ but $a_2 = -a$.

Definition

A function f is said to be **injective** (or **one-to-one** or an **injection**) if

$$f(a_1) = f(a_2) \text{ implies } a_1 = a_2.$$

Example. Determine whether the function $f : \mathbb{Z} \rightarrow \mathbb{Z}$, $f(x) = x + 1$ is injective.

We can show: if $f(a_1) = f(a_2)$, then $a_1 + 1 = a_2 + 1$, then we get $a_1 = a_2$.

Example. $f(x) = x^n$ is injective when n is odd, but is not when n is even.

Definition

A function f is said to be **injective** (or **one-to-one** or an **injection**) if

$$\begin{array}{ccc} p & \xrightarrow{\quad} & q \\ f(a_1) = f(a_2) & \text{implies} & a_1 = a_2. \end{array}$$

Contrapositive form: $q \rightarrow p$, which say: $(a_1 \neq a_2) \text{ implies } (f(a_1) \neq f(a_2))$.
That means: different inputs produce different outputs.

Example. Determine whether the function $f : \mathbb{Z} \rightarrow \mathbb{Z}$,
 $f(x) = x + 1$ is injective.

Example. Determine whether the function

$$f : \{\text{students}\} \rightarrow \mathbb{N}, \quad f(X) = \text{student no. of } X$$

is injective. f is injective because different students should have different student no.

- To show that f is injective, show that if $f(x) = f(y)$,
then $x = y$. p
- To show that f is not injective, find $x, y \in A$ such that
 $x \neq y$ but $f(x) = f(y)$. q

injective: $p \rightarrow q$, where $p: (f(x) = f(y))$ and $q: (x = y)$

not injective: $\sim(p \rightarrow q) = \sim(\sim p \text{ or } q) = (p \text{ and } \sim q)$, which says: $f(x) = f(y)$ but $x \neq y$.

- To show that f is injective, show that if $f(x) = f(y)$, then $x = y$.
- To show that f is not injective, find $x, y \in A$ such that $x \neq y$ but $f(x) = f(y)$.

Example. Determine whether the function $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$ is injective.

Example: $f(x) = x^2$ not injective, we have $f(2) = f(-2)$ but $2 \neq -2$.

Definition

A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is called

- **increasing** if $f(x) \leq f(y)$,
- **strictly increasing** if $f(x) < f(y)$,
- **decreasing** if $f(x) \geq f(y)$, and
- **strictly decreasing** if $f(x) > f(y)$,

whenever $x, y \in \mathbb{R}$ and $x < y$.

Theorem

Strictly increasing and strictly decreasing functions $f : \mathbb{R} \rightarrow \mathbb{R}$ are injective.

Question. How to show $f : \mathbb{R} \rightarrow \mathbb{R}$ strictly increasing/decreasing?

Example. Determine whether the function $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^9$ is injective.

Yes, because x^9 is a strictly increasing function.

Question. How to show $f : \mathbb{R} \rightarrow \mathbb{R}$ strictly increasing/decreasing?

Example. Determine whether the function $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^9$ is injective.

Example. Determine whether the function $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^8$ is injective.

No. From the graph, we see that x^8 is decreasing when $x < 0$, increasing when $x > 0$.

Definition

A function from A to B is called **surjective** (or **onto** or a **surjection**) if

for all $b \in B$, there exists $a \in A$ with $f(a) = b$.

Example. Is the function $f : \mathbb{Z} \rightarrow \mathbb{Z}$, $f(n) = 2n + 1$ surjective?

We ask: for any b in $\mathbb{Z}$, can we always find a in $\mathbb{Z}$, such that $f(a) = 2a + 1 = b$?

Answer: No, because when b is even, it cannot be any $2a + 1$.

So f is not surjective.

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Example. Is the function $f : \mathbb{Z} \rightarrow \mathbb{Z}$, $f(n) = 2n + 1$ surjective?

Example. Determine whether the function

$$f : \overset{\text{finite}}{\{\text{students}\}} \rightarrow \overset{\text{infinite}}{\mathbb{N}}, \quad f(X) = \text{student no. of } X$$

is surjective. No, because if the university has N students, then there is no student with student number $N + 1$.

- To show that f is surjective, show that for every $y \in B$, you can find an $x \in A$ such that $f(x) = y$.
- To show that f is not surjective, show that you can find a particular $y \in B$ such that $f(x) \neq y$ for all $x \in A$.

surjective: $\forall y \in B, \exists x \in A, f(x) = y$.

not surjective: $\sim(\forall y \in B, \exists x \in A, f(x) = y) = (\exists y \in B, \forall x \in A, f(x) \neq y)$.

- To show that f is surjective, show that for every $y \in B$, you can find an $x \in A$ such that $f(x) = y$.
- To show that f is not surjective, show that you can find a particular $y \in B$ such that $f(x) \neq y$ for all $x \in A$.

Example. Is the function $f : \mathbb{R} \rightarrow \mathbb{R}, f(n) = n^2$ surjective?

No because we can find -2 in $\mathbb{R}$ such that $f(x) = x^2 = -2$ has no solution.

- To show that f is surjective, show that for every $y \in B$, you can find an $x \in A$ such that $f(x) = y$.
- To show that f is not surjective, show that you can find a particular $y \in B$ such that $f(x) \neq y$ for all $x \in A$.

Example. Is the function $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(n) = n^2$ surjective?

Example. Is a strictly increasing/decreasing function $f : \mathbb{R} \rightarrow \mathbb{R}$ surjective?

There exists strictly increasing/decreasing function that is NOT surjective.

$f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 2^x$ is strictly increasing,

but it cannot cover the bottom part of the graph (no negative output).

“strictly increasing/decreasing” only works for injective functions.